QUANTITATIVE RESEARCH · SAMPLING & SAMPLING DISTRIBUTIONS · HYPOTHESIS TESTS · CHI-SQUARE · ANOVA

Research Statistics Summary

Everything in your four files — class notes on quantitative research, the Brooks/Cole sampling-distribution deck, and Dr Krishan Pandey's Chi-Square and ANOVA notes — explained from zero. 49 subtopics. Each starts with 'In plain words', then tables, formulas and fully worked examples. Stars show how important a subtopic is for an exam.

★★★★★ must know, asked often ★★★★☆ high ★★★☆☆ medium ★★☆☆☆ low ★☆☆☆☆ skim

‡ rule or number that changes often; check workbook edition

Ch	Chapter	Marks	Priority	Subtopics	5★
1	Research Basics: Quantitative vs Qualitative, Designs, Scales	—	★★★★☆	4	2
2	Sampling Methods	—	★★★★★	5	3
3	Describing Data: Spread, Parameters, Statistics, Estimation	—	★★★★★	4	4
4	Probability Distributions: Binomial, Poisson, Geometric, Normal	—	★★★★☆	4	2
5	Sampling Distributions & the Central Limit Theorem	—	★★★★★	4	4
6	Hypothesis Testing: The Logic	—	★★★★★	5	5
7	Tests for Means: z-test and t-test (worked)	—	★★★★★	7	5
8	Tests for Proportions & Variances (F-test)	—	★★★★☆	3	2
9	Chi-Square (χ^2) Tests — Dr Krishan Pandey	—	★★★★★	4	4
10	Analysis of Variance (ANOVA) — Dr Krishan Pandey	—	★★★★★	4	4
11	Statistical Process Control (control charts)	—	★★★☆☆	3	0
12	Cheat Sheet: Which Test Do I Use?	—	★★★★★	2	2

5★ list (37) — read these first

Ch 1 · Population, sample, sampling frame, sampling error · Scales of measurement

Ch 2 · Simple random sampling · Other random (probability) sampling plans · Non-random (non-probability) sampling

Ch 3 · Measures of spread: range, mean deviation, variance, SD · Parameter vs statistic · Sample SD: why divide by n − 1 · Standard error

Ch 4 · Binomial distribution · Normal distribution & z-score

Ch 5 · What a sampling distribution is · Central Limit Theorem (CLT) · Sampling distribution of the sample mean · Sampling distribution of the sample proportion

Ch 6 · Null and alternative hypotheses · Type I and Type II errors · Decision rules: critical value and p-value · Critical values you must remember · The step-by-step procedure

Ch 7 · Which test: z or t? · One-sample z-test (worked: heights) · One-tailed z-test (worked: production process) · One-sample t-test (worked: bag weights) · Two-sample z-test (worked)

Ch 8 · One-sample proportion test · F-test for equality of variances

Ch 9 · What chi-square is for · Formula & degrees of freedom · Goodness of fit (worked: package colours) · Test of independence (worked: income vs detergent brand)

Ch 10 · What ANOVA does · One-way ANOVA (worked: test anxiety) · Practice problems (answers from the notes) · Two-way ANOVA: main effects & interaction

Ch 12 · Choosing the right test · Ten things beginners get wrong

4★ list (11)

Ch 1 · What 'quantitative' and 'qualitative' research mean · Research designs

Ch 2 · Why the way you choose a sample matters · Observational studies vs experiments

Ch 4 · Poisson distribution

Ch 7 · Finite population correction (worked) · Paired (dependent) t-test

Ch 8 · Two-sample proportion test

Ch 11 · In control vs out of control · $\bar{x}$ chart (for process means) · p chart (for proportion defective)

Sources: your handwritten Quantitative Research notes; 'Sampling Distributions' (Introduction to Probability and Statistics, Brooks/Cole 2006); Dr Krishan K. Pandey, 'Chi Square Test' and 'Analysis of Variance' (OP Jindal Global University). TRAP boxes mark common confusions and places where a source has an error. Lines marked 'Added' fill gaps your notes left blank.

01 Research Basics: Quantitative vs Qualitative, Designs, Scales

≈— marks · 4 subtopics · chapter priority ★★★★☆

What 'quantitative' and 'qualitative' research mean ★★★★☆

In plain words: research is a careful way of answering a question with evidence. **Quantitative** research collects **numbers** (marks, income, ratings out of 5) and analyses them with statistics. **Qualitative** research collects **words and meanings** (interviews, quotations, observations) and looks for themes.

Point (from your notes)	Quantitative	Qualitative
Data form	Numbers	Quotations, text
Time needed	More	Less
Expertise needed	More (statistics)	Less
Typical sample size	At least **200**	Small — about **15 to 20** people
Software mentioned	Statistical software (e.g. SPSS — Added)	**Atlas.ti**
Engagement with respondent	Less (fixed questions)	More (open conversation)
Scope for triangulation	Less	More

Triangulation = checking a finding from two or more sources or methods so that it is more believable

Research designs ★★★★☆

In plain words: a research design is the overall plan of a study — who you study, how, and what you compare.

Approach	Designs listed in your notes	One-line meaning (Added)
Qualitative	Grounded theory	Build a new theory from the data itself
	Ethnography	Live with a group and observe its culture
	Narrative research	Study people's life stories
	Case study	Deep study of one person, firm or event
Quantitative	Experimental	Change one thing (treatment) and measure the effect
	Action research	Researcher solves a real problem while studying it
	Survey	Ask many people fixed questions
	Correlational	Measure whether two variables move together

Population, sample, sampling frame, sampling error ★★★★★

In plain words: the **population** is everyone you want to know about (all students in India). You usually cannot ask all of them, so you pick a smaller group — the **sample**. The **sampling frame** is the list or source from which you can actually reach the population (a college register, a phone directory).

Term	Meaning
Population	Whole group of interest
Sample	Part of the population actually studied
Sampling frame	Where you can access the population data from
Sampling error	Difference between a sample result and the true population value that happens just because you studied a part, not the whole
Non-sampling error	Mistakes not caused by sampling: wrong recording, biased questions, non-response

TRAP Your notes say sampling error is 'any bias attributed to mistake'. Strictly, **SAMPLING ERROR** comes from studying only a sample (chance), while mistakes and bias are **NON-SAMPLING ERRORS**.

Scales of measurement ★★★★★

In plain words: the scale tells you what kind of numbers you have, and so which statistics you may use.

Scale	Meaning	Example	Allowed maths
Nominal	Names/labels only; number has no relation to the object	Gender, brand, colour (1 = red, 2 = blue)	Counts, mode, **chi-square**
Ordinal	Order or rank matters, gaps not equal	1st/2nd/3rd, low/medium/high	Median, ranks
Interval (Added — blank in notes)	Equal gaps, but no true zero	Temperature in °C	Mean, SD, t, z, ANOVA
Ratio (Added)	Equal gaps **and** a true zero	Income, weight, marks	All of the above + ratios

Nominal data are also called **categorical** or **attribute** data (Krishan's chi-square notes)

02 Sampling Methods

≈— marks · 5 subtopics · chapter priority ★★★★★

Why the way you choose a sample matters ★★★★☆

In plain words: a sample is useful only if it looks like the population. If you ask only your friends, you learn about your friends, not about everyone. So statisticians use **random** methods, where chance — not you — decides who is in the sample.

- Parameters (μ, σ, p) describe the population and are usually **unknown**
- Pollster: knows answers are agree/disagree (binomial) but not p, the true share who agree
- Agronomist: knows wheat yield is roughly normal but not μ or σ
- The **sampling plan** (experimental design) decides how much information you get and how reliable it is

Simple random sampling ★★★★★

In plain words: every possible group of n people has the **same chance** of being picked — like drawing names from a hat.

```
89 students, need a group of 5:
1. Number students 01 to 89
2. Read 5 pairs of digits from a random number table
3. If 90-99 or 00 comes up, skip it and read another
4. The 5 students with those numbers form the group
```

Use a computer or random number table

Problems: **non-response**, **undercoverage**, **wording bias**

Observational studies vs experiments ★★★★☆

Type	Meaning	Watch out for
Observational study	Data already existed before you studied it	**Non-response** (only opinionated people reply) · **Undercoverage** (some groups left out) · **Wording bias** (confusing questions)
Experiment	You create data by applying a treatment to units	Population may be hypothetical; sample must be representative and **behave like a random sample**

Other random (probability) sampling plans ★★★★★

Plan	How	Example from the deck
Stratified random	Split population into groups (**strata**), take a simple random sample from **each**	Random sample within every county of California
Cluster	Split into groups (**clusters**, made of neighbouring elements), randomly pick some clusters, study **everyone** in them	Pick 10 random city blocks and interview all residents; pick 10 random counties
1-in-k systematic	Pick a random start among the first k, then every k-th one	Random entry in phone book, then every 50th (1-in-50)

TRAP Stratified = sample **FROM EVERY** group. Cluster = sample **SOME** groups and take **ALL** inside them. Exams swap these.

Non-random (non-probability) sampling ★★★★★

In plain words: here a person, not chance, decides who is included. These are cheap and quick but should **not** be used for statistical inference.

Plan	Meaning
Convenience	Whoever is easy to reach (people walking by)
Judgment (judgement)	Researcher decides who is in
Quota	Sample must match the population on chosen traits (gender, race…)
Purposive (notes)	Chosen for a specific purpose
Snowball (notes)	'Referring' sampling — each respondent refers the next

Notes: probability sampling = simple random, systematic, stratified, cluster

Non-probability = judgment, quota, convenience, purposive, snowball

03 Describing Data: Spread, Parameters, Statistics, Estimation

≈— marks · 4 subtopics · chapter priority ★★★★★

Measures of spread: range, mean deviation, variance, SD ★★★★★

In plain words: the mean tells you the centre. Spread tells you how far the values sit from that centre. If spread (variance) is high, the mean is a poor summary of the data — your notes say 'variance high, mean not good'.

```
Range = largest - smallest        (a crude measure)
Mean deviation = (1/N) Σ |xᵢ - x̄|
Variance σ² = (1/N) Σ (xᵢ - x̄)²     = average squared
distance from the mean
Standard deviation σ = √variance
```

```
Worked: values 1, 2, 3 → mean 2
Variance = [(1-2)² + (2-2)² + (3-2)²] ÷ 3 = (1 + 0 + 1) ÷
3 = 2/3
SD = √(2/3) ≈ 0.816
```

Parameter vs statistic ★★★★★

In plain words: a **parameter** is a number about the whole population (usually unknown). A **statistic** is the same kind of number worked out from your sample (known).

	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
SD	σ_p (σ)	σ_s (s)
Variance	σ_p^2	σ_s^2
Proportion	p (P)	$\hat{p}$ (p)

Going from the known sample to a guess about the unknown population = **estimation**

Hypotheses are tested with sample statistics

Sample SD: why divide by n − 1 ★★★★★

In plain words: if you divide by n when working out a **sample** variance, the answer is on average a bit too small. Dividing by **n − 1** fixes this, so it is 'unbiased'.

```
Population SD:  σₚ = √[ (1/N) Σ(xᵢ - x̄)² ]
Sample SD:      σₛ = √[ (1/(n-1)) Σ(xᵢ - x̄)² ]
With divisor n: E(σₛ²) = ((n-1)/n) σₚ²   → biased (too
small)
Multiply by n/(n-1):  (n/(n-1)) × (1/n) Σ(xᵢ - x̄)² =
(1/(n-1)) Σ(xᵢ - x̄)²  → unbiased
```

Sample (n = 2, with replacement) from {1,2,3}	$\bar{x}$
(1,1)	1
(1,2) (2,1)	1.5
(1,3) (2,2) (3,1)	2
(2,3) (3,2)	2.5
(3,3)	3

Average of all 9 sample means = 2 = μ → **$E(\bar{x}) = \mu$**: the sample mean is an **unbiased estimator**

If σ_p is not given, use σ_s as its estimate

TRAP Your notes use capital N for the sample size in this proof. Read it as small n (sample size).

Standard error ★★★★★

In plain words: if you took many samples, their means would differ a little. The **standard error** is the SD of those sample means. Bigger samples → smaller standard error → more precise.

```
SE of mean = σ / √n
SE of proportion = √(pq / n)
```

σ = SD, σ^2 = variance, $\sigma/\sqrt{n}$ = standard error

04 Probability Distributions: Binomial, Poisson, Geometric, Normal

≈— marks · 4 subtopics · chapter priority ★★★★☆

Binomial distribution ★★★★★

In plain words: you repeat a yes/no trial n times (toss a coin 10 times). Each trial has success chance p and failure chance q = 1 − p. The binomial gives the chance of exactly x successes.

```
P(x) = ⁿCₓ pˣ q^(n−x) ,   x = 0, 1, 2 … n
Mean = np      SD = √(npq)
```

n is finite | Pollster's agree/disagree answers follow a binomial

Poisson distribution ★★★★☆

In plain words: when trials are extremely many (n → ∞) and success is very rare (p → 0) but np stays a fixed number λ, the binomial turns into the **Poisson** — used for counts of rare events (accidents per day).

```
P(X = x) = e^(-λ) λˣ / x! ,   x = 0, 1, 2 … ∞      λ = np
```

Geometric distribution ★★★☆☆

In plain words: you keep tossing a coin and stop the moment you get the first head. The number of tosses needed follows the **geometric** distribution.

```
P(first success on trial x) = q^(x-1) p   (Added)
```

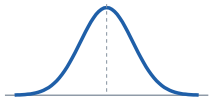
Normal distribution & z-score ★★★★★

In plain words: the normal curve is the symmetric bell shape. Most values sit near the mean; few are far away. A **z-score** says how many SDs a value is from the mean. Converting to z lets one table (Table 3) give every normal probability.

```
z = (x - mean) / SD        → standard normal variate
(mean 0, SD 1)
Runs from -∞ to +∞, centred at 0
```

Area rule (Added)	Share of values
Within ±1 SD	≈ 68%
Within ±2 SD	≈ 95%
Within ±3 SD	≈ 99.7%

Standard normal curve



Centre 0, symmetric

05 Sampling Distributions & the Central Limit Theorem

≈— marks · 4 subtopics · chapter priority ★★★★★

What a sampling distribution is ★★★★★

In plain words: a **statistic** (like the sample mean) changes from sample to sample, so it is a random variable. If you list every value it can take and how often, you get its **sampling distribution**.

```
Population: 3, 5, 2, 1   → all samples of n = 3, without
replacement
3,5,2 → x̄ = 3.33 ; 3,5,1 → 3 ; 3,2,1 → 2 ; 5,2,1 → 2.67
Each x̄ has probability 1/4
```

Found by maths, by simulation, or by theorems like the Central Limit Theorem

Central Limit Theorem (CLT) ★★★★★

In plain words: even if the population is not bell-shaped, the **average** of a large random sample behaves like a bell curve. This is why the normal curve appears everywhere in statistics.

```
If the population has mean μ and SD σ, then for large n:
x̄ ~ approximately Normal(mean = μ, SD = σ/√n)
Sum of n values ~ approximately Normal(mean = nμ, SD =
σ√n)
```

Population shape	How large must n be?
Normal	Any n — x̄ is always normal
Roughly symmetric	Fairly small n
Skewed	n ≥ **30**

Die example (fair die, faces 1–6)	Mean	SD of x̄
n = 1 (flat/uniform)	3.5	1.71
n = 2 (mound-shaped)	3.5	1.71/√2 = 1.21
n = 3 (≈ normal)	3.5	1.71/√3 = 0.987

TRAP The deck says 'toss a fair coin', but the values 1–6 show it is a **DIE** .

Sampling distribution of the sample mean ★★★★★

```
Mean of x̄ = μ        SD of x̄ (standard error) = σ/√n
z = (x̄ - μ) / (σ/√n)
n = 16 from Normal(μ 10, σ 8): P(x̄ > 12) = P(z >
(12-10)/(8/4)) = P(z > 1) = 1 - .8413 = .1587
Soda fills: μ 12.1 oz, σ 0.2, 6-pack: P(x̄ < 12) = P(z <
(12 - 12.1)/(0.2/√6)) = P(z < -1.22) = .1112
```

Sampling distribution of the sample proportion ★★★★★

In plain words: p̂ = x/n is the share of 'successes' in your sample (e.g. share of defective cans). For large n it is also roughly normal.

```
Mean of p̂ = p        SE = √(pq/n)        z = (p̂ - p) /
√(pq/n)
Condition: np > 5 and nq > 5
n = 100, p = .4: P(p̂ > .5) = P(z > (.5 - .4)/
√(.4×.6/100)) = P(z > 2.04) = 1 - .9793 = .0207
Soda: claim p = .05 underfilled, n = 200 (np = 10, nq =
190 → OK)
P(p̂ > .10) = P(z > (.10 - .05)/√(.05×.95/200)) = P(z >
3.24) = 1 - .9994 = .0006 → very unusual if p really is
.05
```

06 Hypothesis Testing: The Logic

≈— marks · 5 subtopics · chapter priority ★★★★★

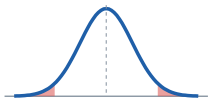
Null and alternative hypotheses ★★★★★

In plain words: a hypothesis test is like a court case. The **null hypothesis (H_0)** is 'nothing special is happening' — the accused is innocent. The **alternative (H_a or H_1)** is what the researcher suspects. We keep H_0 unless the sample gives strong evidence against it.

	H_0 — statistical hypothesis	H_a / H_1 — research hypothesis
Example	μ_girls = μ_boys (no difference in average marks)	μ_girls > μ_boys, or <, or ≠
Always contains	'=' sign	'>', '<' or '≠'

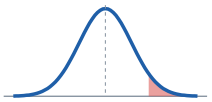
H_a	Test type	Rejection area
μ ≠ value	Two-tailed	Both tails
μ > value	One-tailed (right)	Right tail
μ < value	One-tailed (left)	Left tail

Two-tailed, 5%



2.5% in each tail, z = ±1.96

Right-tailed, 5%



z = 1.645

Type I and Type II errors ★★★★★

Decision	H_0 actually true	H_0 actually false
Reject H_0	**Type I error** (probability α)	Correct (power = 1 − β)
Do not reject H_0	Correct	**Type II error** (probability β)

```
α = P(reject H₀ | H₀ true) = level of significance (LOS)
β = P(do not reject H₀ | H₀ false)
Power of test = 1 - β   → approaches 1 as β → 0
Common α: 1%, 2%, 5%, 10%
```

Neyman–Pearson lemma: fix α, then choose the test with the highest power

ANOVA problem: α = 1% chosen when a false 'drug works' claim would be costly

TRAP Your notes say 'α = consumer risk, β = producer risk'. The standard terms are the other way: **A = PRODUCER'S RISK** (a good lot is rejected), **B = CONSUMER'S RISK** (a bad lot is accepted).

TRAP 'α = 1%' does not mean 'H_0 is true'. It means you accept a 1% chance of rejecting a true H_0 — i.e. 99% confidence.

Decision rules: critical value and p-value ★★★★★

Compare	Decision
|calculated| > |tabulated| (critical)	**Reject H_0**
|calculated| < |tabulated|	Do not reject H_0
p-value ≤ 0.05	**Reject H_0** at 5%
p-value > 0.05	Do not reject H_0 at 5%

Reject H_0 is a strong decision: the sample evidence is statistically significant enough. **Do not reject** is weaker: we only lack evidence; we never 'prove' H_0.

TRAP One line in your notes says 'if p-value > 0.05, reject H_0'. That is wrong. Reject when **P ≤ 0.05** . A small p-value means the data are unlikely if H_0 were true.

Critical values you must remember ★★★★★

Test	Critical value
z, 5%, two-tailed	**1.96**
z, 5%, one-tailed	**1.645**
z, 1%, two-tailed	2.58
z, 1%, one-tailed	2.33 (notes: 2.32)
z, 10%, two-tailed	1.645 (notes round to 1.6)
t, 9 df, 5% two-tailed	2.262
χ^2, 4 df, 5%	9.488
χ^2, 9 df, 5%	16.92
F(2,12), 5%	3.88
F(2,12), 1%	6.93
F(3,16), 1%	5.29

The step-by-step procedure ★★★★★

```
1. State H₀ and Hₐ (decide one- or two-tailed)
2. Set α (usually 5%)
3. Compute the test statistic (z, t, x², F) and its
degrees of freedom
4. Write the decision rule: reject H₀ if statistic ≥
critical value
5. Decide: reject or do not reject
6. State the result in plain English
```

07 Tests for Means: z-test and t-test (worked)

≈— marks · 7 subtopics · chapter priority ★★★★★

Which test: z or t?

★★★★★

In plain words: both compare a sample mean with a value (or with another mean). Use **z** when the sample is large (n ≥ 30) or the population SD is known. Use **t** when the sample is small (n < 30) and you must estimate SD from the sample.

Small sample test (n < 30)	Large sample test (n ≥ 30)
t-test	**z-test** (normal test)
1. One sample (single mean)	1. One sample (single mean)
2. Two independent samples	2. Two samples
3. Paired (dependent) t-test	3. Proportion test

```
One-sample t:  t = (x̄ - μ) / (σₛ/√n)        df = n - 1
One-sample z:  z = (x̄ - μ) / (σₚ/√n)
σ/√n is the standard error (SE)
```

One-sample z-test (worked: heights)

★★★★★

```
n = 400, x̄ = 67.47", μ = 67.39", σₚ = 1.30", α = 5%
I.   H₀: μ = 67.39   vs   Hₐ: μ ≠ 67.39 (two-tailed)
II.  z = (x̄ - μ) / (σₚ/√n)
III. z = (67.47 - 67.39) / (1.30/20) = 0.08 / 0.065 = 1.23
IV.  z tab (5%, two-tailed) = 1.96 → |1.23| < 1.96
V.   Do not reject H₀: the sample is consistent with a population mean of 67.39"
```

If population size is unknown, treat it as infinite

TRAP Your notes say the sample 'belongs to the population whose mean was 67.47'. The population mean is **67.39** ; 67.47 is the sample mean.

Finite population correction (worked)

★★★★☆

In plain words: when the population is small and known (N), and you sample a big part of it, the SE shrinks. Multiply SE by √((N − n)/(N − 1)).

```
z = (x̄ - μ) / [ (σₚ/√n) × √((N - n)/(N - 1)) ]
N = 20, μ = 320, σₚ = 75, n = 5, x̄ = 300
H₀: μ = 320  vs  Hₐ: μ ≠ 320
z = (300 - 320) / [ (75/√5) × √(15/19) ] = -20 / 29.8 = -0.67
|0.67| < 1.96 → do not reject H₀: sample mean consistent with population mean at 95% confidence
```

TRAP One line in your notes writes the correction as (N − n)/(n − 1). The correct factor is **√((N − N)/(N − 1))** , as used in the working.

One-tailed z-test (worked: production process)

★★★★★

```
μ = 50, σₚ = 2.5, n = 12, x̄ = 48.5, α = 5%
H₀: μ = 50 (process unchanged)   Hₐ: μ < 50 (manager guards against a fall)
z = (48.5 - 50) / (2.5/√12) = -1.5 / 0.722 = -2.08
z tab (5%, one-tailed) = 1.645 → |2.08| > 1.645
Reject H₀: the process mean has fallen
```

TRAP The notes write σ/√5 in one line and label H_a as 'H_0: μ < 50'. Use √12 (n = 12) and call μ < 50 the **ALTERNATIVE** .

One-sample t-test (worked: bag weights)

★★★★★

```
Values: 578, 572, 570, 568, 572, 578, 570, 572, 596, 544
(n = 10)
H₀: μ = 578   Hₐ: μ ≠ 578
x̄ = 5720/10 = 572
Σ(x - x̄)² = 36+0+4+16+0+36+4+0+576+784 = 1456
σₛ = √(1456/9) = 12.72
t = (572 - 578) / (12.72/√10) = -1.49
t tab (9 df, 5%, two-tailed) = 2.262 → |1.49| < 2.262 → do not reject H₀
```

```
Second example: n = 12, x̄ = 6576/12 = 548, H₀: μ = 500, Hₐ: μ > 500
σₛ = √(23979/11) = 46.68
t = (548 - 500) / (46.68/√12) = 3.56
t tab (11 df, 5%, one-tailed) = 1.796 (Added) → reject H₀: mean exceeds 500
```

Two-sample z-test (worked)

★★★★★

```
z = [(x̄₁ - x̄₂) - (μ₁ - μ₂)] / √(σ₁²/n₁ + σ₂²/n₂)
Under H₀: μ₁ - μ₂ = 0 → z = (x̄₁ - x̄₂) / √(σ₁²/n₁ + σ₂²/n₂)
σ's unknown → use sample σₛ₁², σₛ₂²
Same σ for both → z = (x̄₁ - x̄₂) / [σ √(1/n₁ + 1/n₂)]  (σ unknown → combined σ)
```

```
n₁ = 100, x̄₁ = 200, n₂ = 150, x̄₂ = 220, population σ = 11, α = 5%
H₀: μ₁ = μ₂ (same population)   Hₐ: μ₁ ≠ μ₂
z = (200 - 220) / [11 √(1/100 + 1/150)] = -20 / 1.42 = -14.08
|14.08| > 1.96 → reject H₀: the two samples come from different populations
```

Paired (dependent) t-test

★★★★☆

In plain words: use it when the same people are measured twice (before and after a course). Work with the differences D.

```
t = D̄ / (σ_D/√n)        σ_D = √[ (1/(n-1)) Σ(Dᵢ - D̄)² ]
df = n - 1
```

08 Tests for Proportions & Variances (F-test)

≈— marks · 3 subtopics · chapter priority ★★★★☆

One-sample proportion test ★★★★★

In plain words: use it for **attributes** — something present or absent (male/female, defective/good). You compare the sample share with a claimed population share.

```
z = (p̂ - P) / √(PQ/n)        Q = 1 - P
Births: n = 3232, boys 1705 → p̂ = 0.5275; claim P = 1/2
z = (0.5275 - 0.5) / √(0.5 × 0.5 / 3232) = 3.13 > 1.96 →
reject H₀: the data do not support a 50:50 ratio
```

```
First class passengers: n = 400, 70 hold first class → p̂
= 0.175; H₀: P = 0.20, Hₐ: P ≠ 0.20
z = (0.175 - 0.20) / √(0.2 × 0.8 / 400) = -0.025 / 0.02 =
-1.25
At 10%: z tab = 1.645 → |1.25| < 1.645 → do not reject H₀
```

```
Defectives: n = 400, 34 defective → p̂ = 0.085; H₀: P =
10%, Hₐ: P < 10%
z = (0.085 - 0.10) / √(0.10 × 0.90 / 400) = -0.015 /
0.015 = -1.0
At 1% one-tailed: z tab = 2.33 → do not reject H₀
```

Two-sample proportion test ★★★★☆

```
z = (p₁ - p₂) / √(P₁Q₁/n₁ + P₂Q₂/n₂)
```

TRAP Your notes write the two-sample proportion test with $\bar{x}$ and σ^2 terms. That is the formula for two **MEANS** . For proportions use p and pq, as above.

F-test for equality of variances ★★★★★

In plain words: it checks whether two samples are equally spread out ('homogeneous'). Homogeneous variance means one group is not more variable than the other.

```
H₀: σ₁² = σ₂²   vs   Hₐ: σ₁² ≠ σ₂²
F = larger sample variance / smaller sample variance
Sample variance = Σ(x - x̄)² / df
Numerator df = df of the larger variance (n₁ - 1);
denominator df = n₂ - 1
```

Also called the **variance ratio test**

Always put the higher variance on top, so F ≥ 1

09 Chi-Square (χ^2) Tests — Dr Krishan Pandey

≈— marks · 4 subtopics · chapter priority ★★★★★

What chi-square is for ★★★★★

In plain words: chi-square works with **counts** in categories (how many chose red, blue…). It asks: are the counts we **observed** close to the counts we would **expect** if nothing special were going on?

Question	Test
Are all package designs / brands equally preferred? (one variable)	**Goodness of fit**
Is income linked to brand chosen? Family size to washing-machine size? Education to job type? (two variables)	**Test of independence** (contingency table)

Variables are **nominal** (categorical / attribute data)

χ^2 is **non-parametric** (distribution-free): no assumption about the population's shape — unlike parametric tests (z, t, F) that assume e.g. normality

χ^2 distribution is skewed for small df and approaches normal as df grows

Notes: χ^2 is a sum of squared standard normal (z) values

Formula & degrees of freedom ★★★★★

```
x² = Σ (O - E)² / E        O = observed frequency, E =
expected frequency
Goodness of fit: df = number of categories - 1
Independence: df = (rows - 1) × (columns - 1)
Expected cell (independence) = (Row total × Column total)
/ Grand total
Reject H₀ if x² calculated > x² critical
```

TRAP Keep expected frequencies with decimals — do not round them (Krishan's notes).

Goodness of fit (worked: package colours) ★★★★★

Colour	O	E	$(O-E)^2$	$(O-E)^2/E$
Red	70	80	100	1.250
Blue	106	80	676	8.450
Green	80	80	0	0.000
Pink	70	80	100	1.250
Orange	74	80	36	0.450
Total	400	400		**11.400**

```
H₀: all colours equally preferred → E = 400/5 = 80 each
x² = 11.40 ; df = 5 - 1 = 4 ; critical at 5% = 9.488
11.40 > 9.488 → reject H₀: colours not equally preferred;
Blue is most preferred → launch blue packaging
```

Test of independence (worked: income vs detergent brand) ★★★★★

Income	Brand 1	Brand 2	Brand 3	Brand 4	Total
Lower	25	15	55	65	160
Middle	30	25	35	30	120
Upper middle	50	55	20	22	147
Upper	60	80	15	18	173
Total	165	175	125	135	600

```
Brands 1, 2 = premium; 3, 4 = economy; stratified random sample of 600
H₀: income and brand preference are independent (no association); α = 5%
Expected (Lower, Brand 1) = 160 × 165 / 600 = 44.0 ; (Upper, Brand 2) = 173 × 175 / 600 = 50.458
Sum of 16 cells of (O - E)²/E = 131.76
df = (4 - 1)(4 - 1) = 9 ; critical at 5% = 16.92
131.76 > 16.92 → reject H₀: brand choice depends on income
```

Biggest contributions $(O-E)^2/E$	Value
Lower × Brand 4	23.36
Lower × Brand 2	21.49
Upper × Brand 2	17.30
Lower × Brand 3	14.08

Conclusion: upper-middle and upper incomes prefer premium brands; lower and middle prefer economy brands → position each accordingly

10 Analysis of Variance (ANOVA) — Dr Krishan Pandey

≈— marks · 4 subtopics · chapter priority ★★★★★

What ANOVA does ★★★★★

In plain words: a t-test compares two means. ANOVA compares **three or more** means at once. It splits the total spread of the scores into two parts: spread **between** the groups (are the group means far apart?) and spread **within** the groups (how much people in the same group differ). If between-group spread is large compared to within-group spread, the groups really differ.

```
F = MSB / MSW
MSB = SSB / df_B      df_B = K - 1        (K = number of groups)
MSW = SSW / df_W      df_W = N_T - K      (N_T = total subjects)
SST = SSB + SSW       df_T = N_T - 1
```

```
SSB = (ΣX₁)²/n₁ + (ΣX₂)²/n₂ + (ΣX₃)²/n₃ - (ΣX_T)²/N_T
SSW = [ΣX₁² - (ΣX₁)²/n₁] + [ΣX₂² - (ΣX₂)²/n₂] + [ΣX₃² - (ΣX₃)²/n₃]
SST = ΣX_T² - (ΣX_T)²/N_T
```

One independent variable → **one-way** ANOVA; two → **two-way** ANOVA

H_0: all means equal; H_a: 'not H_0' — some or all means differ

One-way ANOVA (worked: test anxiety) ★★★★★

	Group 1 (5 h therapy)	Group 2 (10 h)	Group 3 (15 h)
Scores	48, 50, 53, 52, 50	55, 52, 53, 55, 53	51, 52, 50, 53, 50
ΣX	253	268	256
ΣX^2	12,817	14,372	13,114
Mean	50.6	53.6	51.2

```
ΣX_T = 777, N_T = 15, ΣX_T² = 40,303 → (ΣX_T)²/N_T = 40,248.6
SSB = 253²/5 + 268²/5 + 256²/5 - 40,248.6 = 25.2
SST = 40,303 - 40,248.6 = 54.4 ;  SSW = 54.4 - 25.2 = 29.2
MSB = 25.2/2 = 12.6 ; MSW = 29.2/12 = 2.43 ; F = 5.178
F.05(2,12) = 3.88 → 5.178 ≥ 3.88 → reject H₀
```

Source	SS	df	MS	F	p
Between	25.20	2	12.60	5.178	< .05
Within	29.20	12	2.43		
Total	54.40	14			

Result: amount of therapy significantly affects test anxiety scores

Next: **post hoc** (after-the-fact) comparisons to see **which** pairs differ — only when F is significant

Practice problems (answers from the notes) ★★★★★

Problem	F (df)	F crit	Decision	Meaning
1. Antidepressant: placebo vs low vs moderate dose (Beck scores)	11.26 (2,12)	6.93 at 1%	Significant, p < .01	Placebo differs from both doses; low vs moderate no difference → low dose recommended
2. US history scores by major (4 groups, 32 students)	0.04 (3,28)	—	Not significant	All majors equal (equally poor); if wrong, it's a Type II error
3. Rat environment and maze learning (4 groups)	12.71 (3,16)	5.29 at 1%	Significant, p < .01	Differences appear only between groups at least two levels apart

Problem 1: α = .01 chosen because a false 'drug works' claim (Type I) would be costly

Problem 2: α = .05 because little risk either way

TRAP In Problem 3 the notes say enriched vs super enriched is 'no significant difference' but print 'p < .05' beside it. Treat it as **NOT SIGNIFICANT** .

Two-way ANOVA: main effects & interaction ★★★★★

In plain words: two factors at once (e.g. protein level and gender). A **main effect** is the effect of one factor on its own. An **interaction** means the effect of one factor changes depending on the other.

Protein breakfast study	SS	df	MS	F	Result
Protein level	20	1	20.00	8.89	Significant, p < .01
Gender	45	1	45.00	20.00	Significant, p < .01
Protein × Gender	5	1	5.00	2.22	Not significant
Within	36	16	2.25		
Total	106	19			

Means: males high 8, low 5; females high 4, low 3

High protein → better fitness; males score higher

Life satisfaction study	SS	df	MS	F	Result
Age	180	2	90.00	49.09	Significant, p < .01
Gender	30	1	30.00	16.36	Significant, p < .01
Age × Gender	0	2	0.00	0.00	No interaction
Within	44	24	1.83		
Total	254	29			

Older adults most satisfied, young least; women more satisfied than men

Without post hoc tests you can't say exactly which age groups differ

11 Statistical Process Control (control charts)

≈— marks · 3 subtopics · chapter priority ★★★☆☆

In control vs out of control ★★★★☆

In plain words: a factory process always varies a little by chance (**random variation**). If something specific goes wrong (a machine slips), that is an **assignable cause** — it can be found and fixed. A process with only random variation is **in control**.

Out of control → reduce variation and bring the measurements back within limits

x̄ chart (for process means) ★★★★☆

```
Take k samples of size n; x̿ = grand mean of all nk
values; s = SD of all nk values
Centreline = x̿
LCL = x̿ - 3s/√n        UCL = x̿ + 3s/√n
A sample mean outside the limits → process may be out of
control
```

Based on the CLT: almost all x̄ fall within μ ± 3σ/√n

p chart (for proportion defective) ★★★★☆

```
p̄ = Σp̂ᵢ / k   (average of the k sample proportions)
Centreline = p̄
LCL = p̄ - 3√(p̄(1 - p̄)/n)        UCL = p̄ + 3√(p̄(1 - p̄)/n)
A sample proportion outside the limits → process may be
out of control
```

12 Cheat Sheet: Which Test Do I Use?

≈— marks · 2 subtopics · chapter priority ★★★★★

Choosing the right test ★★★★★

Your question	Data	Test	Statistic
Is one mean equal to a value? (n ≥ 30 or σ known)	Numbers	One-sample z	$z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$
Same, but n < 30 and σ unknown	Numbers	One-sample t	$t = (\bar{x} - \mu)/(s/\sqrt{n})$, df n − 1
Do two group means differ? (large)	Numbers	Two-sample z	$(\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
Same people measured twice?	Numbers	Paired t	$\bar{D}/(s_D/\sqrt{n})$
Is a share equal to a claimed value?	Yes/no	One-sample proportion z	$(\hat{p} - P)/\sqrt{(PQ/n)}$
Do two variances differ?	Numbers	F-test	larger s^2 / smaller s^2
Do 3+ group means differ?	Numbers	ANOVA	F = MSB/MSW
Do counts fit an expected pattern?	Categories	χ^2 goodness of fit	$\Sigma(O - E)^2/E$, df k − 1
Are two categorical variables linked?	Categories	χ^2 independence	$\Sigma(O - E)^2/E$, df (r−1)(c−1)

Ten things beginners get wrong ★★★★★

- Reject H_0 when p is **small** (≤ α), not large
- 'Do not reject' ≠ 'H_0 proved true'
- Type I = rejecting a true H_0 (α); Type II = keeping a false H_0 (β)
- Sample SD divides by **n − 1**; population SD by N
- Standard error = σ/√n — it shrinks as n grows
- Stratified samples every stratum; cluster takes all members of some clusters
- χ^2 needs counts (nominal data), not means; it is non-parametric
- ANOVA tells you **that** means differ; post hoc tests tell you **which**
- Two-tailed 5% uses 1.96; one-tailed 5% uses 1.645
- CLT: sample means are roughly normal when n ≥ 30, even from a skewed population